Implementation Differential and Provenance Supplement

From the 2023 article (MATLAB era) to the 2026 Python engine

Companion to the didactic manual (`docs/manual.pdf`) — repository `big.bi-supplement`, commit `4a0d76290aee`, DOI 10.5281/zenodo.23239483

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Documentation supplement — no engine code is changed.

Contents

1 Purpose and reading guide **1**

2 Provenance and versions **2**

3 The model as published (2023 reference) **3**

4 The differential **4**

5 Focus: word-onset anchoring across a pause **7**

6 Why the published demonstrations are nevertheless reproduced **9**

7 Corrigenda to the manual **10**

8 Version and provenance statement **12**

9 Reproducibility **12**

1 Purpose and reading guide

The didactic manual of this repository explains how to reproduce the demonstrations of Berthommier (2023) with the Python engine shipped in `big.bi-supplement`. This supplement answers the question that precedes reproduction: *which* parameter values and assembly rules implemented in the 2026 engine differ from those published in the 2023 article, *where* each difference lives in the code, *what* it changes numerically in the simulations, and *why* the published demonstrations are nevertheless reproduced. It is the documentation-level fix recommended by the independent compliance audit [AUD]; it proposes and implies no engine code change.

Every factual claim carries a source tag:

- **[A §x]** — the 2023 article, F. Berthommier, "Why can big.bi be changed to bi.gbi? A mathematical model of syllabification and articulatory synthesis", arXiv:2307.02299 (Interspeech 2023), section §x;
- **[M l.n]** — the didactic manual, `docs/manual.tex` (29-page PDF at the same commit), line n;

- **[R file:line]** — the repository engine or scripts at commit `4a0d76290aee`;
- **[AUD]** — the 104-test compliance audit report (Oct. 2026);
- **[D5]** — the difference-5 effect report (Oct. 2026), the follow-up experiment summarized in §5 below;
- **[EXT §x]** — the external comparative audit of the consonant selection vectors $C \in \{\text{b, d, g}\}$ in the two original engines, October 13, 2026, reference [6].

Section §7 issues three *corrigenda*: two factually wrong statements of the manual, superseded by the replacement texts given there, and one family of wrong labels in the demonstration figures generated by the repository's own figure generator. No other part of the manual is affected. The intended audience is researchers who reproduce, extend, or cite the demonstrations.

Two conventions. First, Maeda parameter indices are 1-based in the article and the manual and 0-based in the code; article $\{1, 2, 6\}$ = code $\{0, 1, 5\}$ (Jaw, Body, LipH), and the seven parameters are Jaw, Body, Dorsum, Tip, LipP, LipH, Hy [R `synthSYL/constants.py:19`]. Second, magnitudes are expressed in *Maeda units* (M.u.), the natural units of the 7-parameter articulatory flow $P(t)$.

2 Provenance and versions

Lineage (author-asserted). The 2023 article's demonstrations were produced with the author's MATLAB implementation. The engine shipped in this repository is Python: it descends from the author's `timit-to-Maeda` package ("the original synthSYL") and was reworked during 2026 to produce every demonstration presented here — videos, HTML simulations, article-format figures, and the manual itself. The MATLAB source is not part of the audited corpus, and this supplement makes no claim about its internals; everything published is attributed to the article. The verifiable axis of this document is therefore: *values and rules published in the article (2023 era)* ↔ *values and rules implemented in the engine (2026 era)* [AUD; R].

What "verbatim from the original code" means. The engine registry opens with this statement:

`synthSYL/constants.py:6-8` (verbatim)

> "All magic numbers from the original monolithic plugin are replaced by descriptively-named constants here. Every value is taken verbatim from the original code so that numerical identity is preserved."

"The original code" designates the pre-existing Python engine (`timit-to-Maeda`), *not* the article. Registry inheritance therefore guarantees numerical identity with the Python lineage — and that lineage does not coincide everywhere with the values printed in the article. This ambiguity is one of the gaps the present supplement resolves. Throughout §4, every non-article engine value is labelled either *registry inheritance* (carried over verbatim from `timit-to-Maeda`, per the quote above) or *author ruling (dated)*. Two rulings are documented in the code: the word-onset anchoring rule, author-validated 2026-10-06 [R `synthSYL/gesture.py:399-406`], and `VALRECT` = 1.10, ruled 2026-10-07 [R `config.py:43-46`].

The 2026 rework, as documented. The README section "Key Modifications to the Original synthSYL" lists four items [R README]: (1) `delta_o` and `delta_e` added as pipeline kwargs; (2) the syllable-onset anchor at a '.' C.C boundary weighted by `COEFCEN` (the previous vowel weighted by the chaining coefficient); (3) the Vo formula corrected to $\delta_o \cdot \rho_V$ (matching the article); (4) non-initial word onsets anchored on the previous Ve. Items 1 and 3 conform to the article's formulas [A §2.2]; item 2 is the inter-syllable chaining mechanism of the demonstrations; item 4 is difference 5

of §4 below — the README documents the rule and its motivation, but not its divergence from a literal reading of the article nor the conditions under which the two readings coincide (§5).

Version pinning. Repository `big.bi-supplement`, commit `4a0d76290aee`, MIT license, DOI 10.5281/zenodo.23239483 [R]. The manual (`docs/manual.pdf`, 29 pages) and all shipped artifacts were produced at this commit.

Summary statement [AUD]

The engine faithfully implements the article's equations, but its parameters are not exactly those of the 2023 article; the differences are catalogued in §4, and none of them is an equation-level change.

3 The model as published (2023 reference)

The model plans formant trajectories directly in the complex plane. Targets are polar points (ρ, θ); the *coordination function* (article Eq. 1) maps a target to the 7 Maeda parameters,

$$P_i \;=\; \Omega_i \;+\; \rho \cdot \Psi_{1,i} \cdot \cos(\Psi_{2,i} - \theta)\,, \tag{1}$$

where Ω, Ψ_1, Ψ_2 are per-parameter calibration constants (article Table 1) [A §2.1]. Motion between two targets follows a circular arc in the plane (article Eqs. 2–3) with curvature K and the cosine velocity profile $\rho(t) = \cos(\theta/2)$ with $P_{\text{exp}} = 1$ [A §2.1]. Superposition (article Eq. 4) is selection-based: at every instant each segment activates a consonant set S_c and a vowel set S_v of the 7 Maeda parameters [A §2.1–2.2].

Published settings [A §2.1]: $K_{\text{voy}} = 30$ (vowel arcs), $K = 10$ (consonant arcs), $T = 16$ steps per syllabic period in didactic mode (1 step = 10 ms, so $T = 160$ ms), $\nu = -1$, $P_{\text{exp}} = 1$.

Anchoring vowels and pauses [A §2.2]:

Article §2.2 — anchoring vowels (Q1, verbatim [A])

"When they are absent, they can be far from the neutral vowel (with $\rho = 0$ and $P = \Omega$) and they have the same θ as the closest vowel but ρ is multiplied by specific coefficients δ_o and δ_e (which will be set later). Thus, we have $V_o = (\delta_o.\rho_V, \theta_V)$ and $V_e = (\delta_e.\rho_V, \theta_V)$."

Article §2.2 — pause (Q2, verbatim [A])

"the pause is defined as the lack of concatenation and coarticulation between two syllables. This is consistently defined as a diphthong transition between the previous V_e and the next V_o with an arbitrary duration T_p."

The article's demonstrations use $\delta_o = \delta_e = \delta$ throughout, with three published settings: $\delta = 0.5$ (salient schwa), $\delta = 0.7$ (audible schwa), and $\delta = 1$ (fusion, article Figure 4) [A §2.2, §4]:

Article §4 — audible schwa (Q3, verbatim [A])

"the graphs of /big/ and /bi/ are concatenated with $\delta_o = \delta_e = 0.7$ to insert an audible schwa-like vocoid between the two consecutive consonants."

Published consonant positions and selections [A §3]: /b/ at (ρ_b=1, θ_b=$\pi/3$) with $S_c = \{1, 2, 6\}$; /d/ around (ρ_d=1.2, θ_d=$3\pi/2$) with $S_c = \{1, 2, 3, 4\}$; /g/ velar at $(1.2, \pi/3)$ and palatal at $(1.1,\ 23\pi/12)$ with $S_c = \{1, 2, 3, 4\}$ (the /g/ is velar for back vowels, palatal for front vowels; the /d/ position depends slightly on the following vowel). Corner vowels are published at $\rho = 1$, and the calibration method is stated openly: the model's parameters "can be adjusted manually

by simple hearing" [A §1], and "[t]he tuning of /b,d,g/ has been realised by hearing at first" [A §3]. For consonant clusters:

Article §3 — cluster selection (verbatim [A]; checked against the arXiv rendering for this supplement)

"[T]he tuning of the six pair clusters of /b,d,g/ follows the same principle, but there is a special grouping of articulators when /b/ is involved in the pair. In this case $S_c = \{1, 2, 3, 6\}$ *and otherwise this is* $S_c = \{1, 2, 3, 4\}$*."*

Published demonstration [A §4]: `big.bi` transforms into `bi.gbi` as $\delta \to 1$; the gain of consonant clustering is one period $T = 160$ ms. The article's supplement demonstrates `ib ib` → `bi bi`. Article Figure 3 reports nominal locus slopes ≈ 0.70 (/b/), 0.55 (/d/), 2.0 (/g/ velar), and 0.75 (/g/ palatal).

4 The differential

Table 1 is the central exhibit of this supplement: one row per difference between the article-published values/rules and the engine at commit `4a0d76290aee`. Row 1 carries two entries (1 and 1b): both lie in the Tip row of the coordination matrix. Paragraphs after the table give each row's origin, effect, and current documentation status. The two manual statements quoted in §7 are superseded by the corrigenda there.

Row 1/1b — coordination matrix, Tip row (registry inheritance). The engine stores each row as $[\Psi_1, \Omega, \Psi_2]$ [R constants.py:21–33]; the Tip row is $[-2.75,\ 0.5,\ \pi]$ [R constants.py:28], against the article's $\Omega = 0$, $\Psi_1 = -3$ (Table 1) [A]: *both* constants differ. Quantified by the audit: up to 0.70 M.u. on Tip for /d/ and /g/ (measured on the arc /d/→/a/), and zero for /b/ and /gb/, whose selection vectors exclude Tip [AUD]. This row is the audit's single non-conformity, of severity below P0 [AUD]. Substituting the article's Table 1 matrix into the engine (a counterfactual run) yields identity on all seven parameters [AUD], so no other row deviates. Documentation status: omitted — the manual asserts a different (single-entry) difference; superseded by Corrigendum 1 (§7).

Row 2 — dental angle (registry inheritance). `THETA_DENTAL` = $23\pi/16$ versus the article's $3\pi/2$: $\Delta = \pi/16 = 11.25°$ [R constants.py:46; A §3]. The locus and trough demonstration scripts explicitly restore the article value before measuring [R `scripts/run_locus_trough_demos.py:206-216`], so the published-slope demonstrations are computed at the article's angle (§6). This difference is absent from the manual and from code comments.

Row 3 — radii (recalibration by ear; registry inheritance). Engine: $\rho_b = 1.2$ [R constants.py:74], /i/ = 0.9 [R constants.py:77], /a/ = 0.8 [R constants.py:79]; article: $\rho_b = 1$, corner vowels $\rho = 1$ [A §1, §2.1, §3]. The divergence is consistent with the article's own calibration method ("by simple hearing" [A §1]): the 2026 registry is a later state of the same ear-tuned constants. Quantified: the manual's trough ablations measure the /ibi/ Body peak at +1.50 with $\rho_b = 1.2$ and +1.25 with the article's $\rho_b = 1.0$ (plateau −2.25 in both cases) [M l.870–872; M l.1148–1154]. Documentation status: ρ_b is documented (the demo scripts set it explicitly and the manual discusses both values); the vowel radii (0.9 and 0.8 vs published 1) are not flagged as deviations — the manual states them without comment [M l.276–277].

Table 1: Master differential — article (2023) vs engine (2026), commit `4a0d76290aee`. Indices are 1-based in article numbering unless a code vector is shown. M.u. = Maeda units. Code paths are relative to the repository root: `constants.py`, `phonology.py`, `gesture.py`, `trajectory.py` and `polar.py` are under `synthSYL/`; `config.py` is at the root.

#	Item	Article (2023)	Engine (2026)	Code [R]	Quantified effect	Doc. status
1/1b	CO Tip row (Ω, Ψ_1)	$\Omega = 0$; $\Psi_1 = -3$ (Table 1)	$\Omega = 0.5$; $\Psi_1 = -2.75$ (registry inheritance)	constants.py: 28	up to 0.70 M.u. on Tip for /d/, /g/ (arc /d/→/a/); zero for /b/, /gb/ [AUD]	omitted — manual states a different claim (Corr. 1)
2	θ_{dental}	$3\pi/2$ (§3)	$23\pi/16$ ($\Delta = \pi/16 = 11.25°$)	constants.py: 46	reproduction scripts restore the article value (§6)	absent
3	Radii	$\rho_b = 1$; corner vowels $\rho = 1$ (§1, §2.1)	$\rho_b = 1.2$; /i/ = 0.9; /a/ = 0.8 (recalibration by ear)	constants.py: 74, 77, 79	Body peak +1.50 vs +1.25 at $\rho_b = 1.0$ (manual ablations) [M l.870–872, l.1148–1154]	ρ_b documented; vowel radii not flagged [M l.276–277]
4	Cluster selection S_c	/b/-clusters → $\{1, 2, 3, 6\}$; otherwise $\{1, 2, 3, 4\}$ (§3)	place-index sum (b=0, d=1, g=2): $\leq 2 \rightarrow \{1, 2, 3, 6\}$; $> 2 \rightarrow \{1, 2, 3\}$ (article numbering; verified by execution)	phonology.py: 337–341; constants.py: 89–92	4 of 9 /b,d,g/ pairs diverge (dd, dg, gd, gg); /gb/ conforms [AUD]	wrong — manual documents the article's rule (Corr. 2); ancestral [EXT]
5	Word-onset anchoring	literal reading: $V_{o,2} = (\delta_o\, \rho_{V_2}, \theta_{V_2})$ (§2.2)	$V_{o,2} := (\delta_e\, \rho_{V_1}, \theta_{V_1})$ (author ruling 2026-10-06)	gesture.py: 385–432	0 (bitwise) on all 6 published configurations; up to 2.0812 M.u. outside them [D5]	partial (README item 4)
6	VALRECT (soft rectification)	not published	1.10 (historical timit-to-Maeda value 0.75; author ruling 2026-10-07)	config.py: 43–46	locus-slope effect ≤ 0.23 (/g/ velar), ≤ 0.014 (/b/, /d/) [AUD]	well documented
7	Structural conventions (family)	not described as such	opint = −1 (first sample dropped at releases); plateaus as constant blocks; terminal anchors $[0, 0]$; parametrised T_p; 3-consonant extension; display arcs $\cos^2(\theta/2)$; F2 offset 3 steps	polar.py: 43–51; trajectory.py: 56; gesture.py: 487–491; config.py: 47–49	joint sampling jump 0.41 M.u. (analytic, $D = 16$); display-arc gap ≈ 0.12 [AUD]	partial (display side in manual §6; engine side not presented as conventions)

Note (row 3): further registry values have *no* article counterpart, so they cannot differ from a published value — the cluster-context angles `THETA_CC_D` $= -2.5\pi/6$ and `THETA_CC_G` $= -\pi/12$ and the cluster-velar radius `RHO_CLUSTER_VELAR` $= 1.1$ [R constants.py:53–55, 81].

Row 4 — cluster selection (registry inheritance). Engine rule: for a two-consonant cluster, sum the place indices of the two consonants (b, p, m, f, v = 0; d, t, n, s, z = 1; g, k = 2; ...) and select $\{0,1,2,5\}$ when the sum is ≤ 2, $\{0,1,2\}$ when it is > 2 [R phonology.py:133–141, 337–341; constants.py:89–92]. Executed on the nine /b,d,g/ pairs for this supplement, the rule returns $\{1,2,3,6\}$ (article numbering) for /bb/, /bd/, /bg/, /db/, /dd/, /gb/ and $\{1,2,3\}$ for /dg/, /gd/, /gg/ [R]. The article's rule is /b/-involvement (§3, quote above): $\{1,2,3,6\}$ when /b/ is in the pair, $\{1,2,3,4\}$ otherwise [A §3]. Divergence: 4 of the 9 pairs (/dd/, /dg/, /gd/, /gg/); /gb/ — the cluster of the article's demonstration — conforms; among the six pairs listed in the article, /dg/ and /gd/ diverge [AUD]. The singleton selections do conform (/d/ and /g/ activate $\{1,2,3,4\}$) [R phonology.py:303–304; A §3]. Documentation status: wrong in the manual — the Demo 4 table and rule paragraph present the article's rule as the engine's; superseded by Corrigendum 2 (§7).

Row 4a — lineage of the cluster rule (external evidence) [EXT]. The external note [6] compares the two original engines, `timit-to-Maeda` and `Syllable_Synthesis`; it never opens the article and never tests this repository. Its findings: the isolated-consonant selectors "are identical after index conversion" [EXT §2.3]; the binary-cluster masks are identical — art1 rows $\{0,1,2,5\}$ / $\{0,1,2\}$ (the duplicated Jaw entry is inert) coincide with `timit-to-Maeda`'s `ART_CLUSTER` — "The masks are therefore identical" [EXT §3.1]; the index rule is the same in both engines, `art_idx = 1[idx_c1 + idx_c2 > 2]` — "The decision rule is the same" [EXT §3.2], with the same partition {bb, bd, db, bg, gb, dd} vs. {dg, gd, gg}; the engines differ only in trigger logic (enumerated syllable patterns vs. generic $m \in \{2,3\}$), in three-consonant support, and in polar-target formulas for /d/ and /g/ [EXT §9].

Chain of custody. The note's `timit-to-Maeda` excerpts (`_CONSONANT_ENTRIES`, `ART_CLUSTER`, `cluster_art_params`) match this repository byte-for-byte at `constants.py:89-92` and `phonology.py:337-341`, :173–180, modulo one cosmetic character (uvular key: IPA turned-r glyph in the repository, ASCII R in the excerpt). The engine-versus-engine identity therefore extends to the 2026 repository at the selector layer.

Genealogical conclusion. Because the threshold rule and both masks are shared by the two 2023 engines, the non-article selector rule is *ancestral*: it belongs to the 2023 lineage, and the article's /b/-conditional rule was never implemented in any engine of the family. The 2026 rework did not introduce it.

Reserve on the note. Its two "residual differences" for the isolated /d/ and /g/ polar targets are identical modulo 2π ($-9\pi/16 + 2\pi = 23\pi/16$; $-\pi/12 + 2\pi = 23\pi/12$) — a notational, not numerical, difference [EXT §5].

Row 4b — scope of the divergence. *Results cited in the article: none affected.* (i) Isolated consonants /b/, /d/, /g/ (§3 of the article; its Figures 1 and 3): the engine's selectors conform — /b/ $\to \{1,2,6\}$; /d/, /g/ $\to \{1,2,3,4\}$ [R phonology.py:173–180, :303–304]. (ii) The article's CC-pair text (§3) is descriptive only: it publishes no number and no figure for the pairs. (iii) §4 and Figure 4 (`big.bi` ↔ `bi.gbi`): the only cluster the article uses is /gb/, with $S_c = \{1,2,3,6\}$ — conform, and validated end-to-end by the audit (cluster pair /gb/: test C.4b $\leq 2.2 \cdot 10^{-15}$; Eq. (4) end-to-end $\leq 4.4 \cdot 10^{-15}$) [AUD]. The intersection of the divergent pairs {dd, dg, gd, gg} with the article's cited results is empty.

Extended CC simulations of the supplement: affected. Demo 5 synthesizes six CVCCV utterances abda, abga, adba, adga, agba, agda [R run_article_demos.py:296–304]. Exactly two run on a divergent pair: `adga` (/dg/) and `agda` (/gd/) — synthesized with $S_c = \{1,2,3\}$ instead of the

announced $\{1, 2, 3, 4\}$, and their published panel labels display the wrong vector (Corrigendum 3). All other demonstrations (`big.bi`, `bi.gbi`, Demo 1 /ibia/, locus runs) are untouched.

Row 5 — word-onset anchoring across a pause (author ruling 2026-10-06). Treated in depth in §5. Effect: exactly 0 (bitwise) on all six published configurations; up to 2.0812 M.u. outside them [D5] — the largest quantified deviation of the audit. Documentation status: partial — README item 4 and the in-code docstring state the rule and its motivation; the divergence from a literal reading of the article and the conditions under which the two readings coincide are stated for the first time in this supplement.

Row 6 — VALRECT, soft rectification (author ruling 2026-10-07). The article publishes no such setting; the engine applies

`config.py:43-46` (comment; line breaks removed)

```
VALRECT = 1.10 # VLAM soft-rectification (author ruling 2026-10-07, reduces the
perceived fizz on /i/; historical canonical value from timit-to-Maeda:  0.75; applied
to the area function, not to articulatory params)
```

Quantified by the audit: locus-slope effect ≤ 0.23 (/g/ velar) and ≤ 0.014 (/b/, /d/) [AUD]. Well documented: the config comment, manual §7.2 [M l.721–726] and the Demo 2 note [M l.967–968], and the README headline table. One refinement: those documents call 0.75 "the article's canonical value"; per the audit the article publishes no VALRECT — 0.75 is the historical `timit-to-Maeda` default, which the 2026 ruling raises to 1.10 [AUD; R config.py:43–46].

Row 7 — structural conventions (family; registry inheritance). Not differences of published values but conventions of the Python engine that the article does not describe as such: the first sample of each block is dropped at releases (interpolation mode `opint` $= -1$) [R polar.py:43–51; projection.py:45–52]; vowel and consonant plateaus are materialized as constant blocks (`np.tile`) [R trajectory.py:56]; utterance edges carry terminal anchors $[0, 0]$ [R gesture.py:487–491]; pause durations T_p are parametrised; a three-consonant cluster extension exists with no article counterpart [R phonology.py:342–352]; the polar *display* draws arcs with a $\cos^2(\theta/2)$ profile while the engine uses $\cos(\theta/2)$ [R polar_primitives.py; M §6]; and the locus measurement samples F2 30 ms after release (`LOCUS_OFFSET_STEPS` $= 3$) [R config.py:47–49]. Joint effect of the sampling-side conventions: an analytic 0.41 M.u. jump at block joints ($D = 16$); display-vs-engine arc gap ≈ 0.12 M.u. [AUD]. Documentation status: partial — the display conventions are documented (manual §6, "Polar Trajectory Reconstruction"); the engine-side conventions (plateau blocks, `opint`, the 0.41 M.u. jump) are not presented as conventions.

Ranking at trajectory level. Outside the published grid, the differences rank: word-onset rule 2.08 > CO Tip row 0.70 > joint sampling jump 0.41 > display-arc gap 0.12 M.u. [AUD; D5]. Inside the published grid, rows 1, 3, 6 and 7 act at their quantified magnitudes while row 5 is exactly zero (§5).

5 Focus: word-onset anchoring across a pause

The two readings. The article defines the pause as a diphthong transition between the previous V_e and the next V_o (Q2), with $V_o = (\delta_o \rho_V, \theta_V)$ anchored on "the closest vowel" (Q1) [A §2.2]. For a non-initial word that begins with a consonant after a pause, two readings of "closest vowel" — and of the scaling coefficient — are possible:

Engine rule (2026) [R gesture.py:385–432]

The onset anchors on the *previous* word's V_e point, scaled by the *word* coefficient δ_e:

$$V_{o,2} := (\delta_e \cdot \rho_{V_1},\ \theta_{V_1})$$

Article-literal reading (2023) [A §2.2]

$$V_{o,2} \ = \ (\delta_o \cdot \rho_{V_2},\ \theta_{V_2})$$

`synthSYL/gesture.py:399-406` — docstring (Q6, verbatim)

"Non-initial word (after a pause): the pause is the diphthong transition between the PREVIOUS V_e and the next V_o (article §2.2), so the onset anchors on the previous word's V_e point — $V_o := (\delta_e \cdot \rho_{\text{prev}}, \theta_{\text{prev}})$. Consequences: with COEFCEN = 1 ($\delta_e = 1$, $V_e = V$) the word-2 onset departs from the FULL vowel and there is no return to the half-radius schwa at the word boundary; with the symmetric $\delta_o = \delta_e$ the value is unchanged (0.45 for /i/), preserving the 'big@bi' schwa demonstrations. (Author-validated 2026-10-06.)"

Coincidence lemma. The two readings produce the same anchor if and only if

$$\delta_e \, \rho_{V_1} \ = \ \delta_o \, \rho_{V_2} \qquad \text{and} \qquad \theta_{V_1} \equiv \theta_{V_2} \pmod{2\pi}. \tag{2}$$

Corollary. At the published conditions — the article always sets $\delta_o = \delta_e = \delta$, and the demonstrations place the same vowel on both sides of the pause — both conditions hold identically, so the two readings coincide exactly. For `bi bi` with the engine's /i/ ($\rho = 0.9$, $\theta = 5\pi/3$): both readings give $V_{o,2} = (0.45, 5\pi/3)$ at $\delta = 0.5$, $(0.63, 5\pi/3)$ at $\delta = 0.7$, $(0.90, 5\pi/3)$ at $\delta = 1.0$ [R; M §4].

Structural argument. The single-word utterances `big.bi` and `bi.gbi` contain no non-initial word onset at all: their internal '.' boundary is an inter-syllable boundary governed by the chaining rule (README item 2, the `COEFCEN`-weighted '.' C.C anchor), an assembly rule identical under both readings. The number of divergent application sites in these demonstrations is therefore zero for structural reasons, independently of the lemma [D5].

The D5 experiment. Each case was run twice through the complete pipeline, identical except for `_insert_vowel_onset_anchors` (engine rule vs literal rule); the table reports the maximum absolute difference of the 7-parameter flow [D5]:

Table 2: Difference-5 experiment: engine rule vs article-literal rule at the published conditions ($\delta_o = \delta_e = \delta$) and at one control outside them. max $|\Delta P|$ over the 7 Maeda parameters and the whole utterance [D5].

Utterance	δ	Onset sites	Divergent branch	max $\lvert\Delta P\rvert$ (M.u.)	Reading
bi bi	0.5	2	1 (non-initial word)	0 (exact, bitwise)	coincidence (lemma (2))
bi bi	0.7	2	1 (non-initial word)	0 (exact)	coincidence (lemma)
bi bi	1.0	2	1 (non-initial word)	0 (exact)	coincidence (lemma)
big.bi	0.7	1	0	0 (exact)	structural (no site)
big.bi	1.0	1	0	0 (exact)	structural (no site)
bi.gbi	1.0	1	0	0 (exact)	structural (no site)
control: "big ba"	δ_o=0.5, δ_e=0.7	2	1	**2.0812**	divergence

The control in detail. For `big ba` with $\delta_o = 0.5$, $\delta_e = 0.7$, the two anchors split: the engine rule gives $(0.63, 5\pi/3)$ — $\delta_e \cdot 0.9$ on the previous /i/ — while the literal rule gives $(0.40, \pi)$ — $\delta_o \cdot 0.8$ on the word's own /a/; the polar gap is $\Delta\rho = 0.23$ and 1.73 rad on the unit circle (matching audit test D.6 [AUD]). Per-parameter maxima: Body 2.0812 (at $t = 129$ steps ≈ 1.29 s), Tip 1.99, LipH 1.81, Jaw 1.08, while Dorsum, LipP and Hy remain at 0.23–0.35; word 1 is untouched ($\Delta P = 0$ for $t < 112$ steps) [D5]. The four strongly affected parameters are the consonant selection of the /b/, $S_c = \{1, 2, 6\}$ = Jaw, Body, LipH, together with Tip from the vowel selection $S_v = \{3, 4, 5, 7\}$ = Dorsum, Tip, LipP, Hy — consistent with the split of article Eq. 4 [D5].

Interpretation. Within the published grid the rule is invisible: Table 2 shows exact zeros for all six published configurations, and the README records that "all reference reproductions remain bit-identical" (Q10, quoted in §6). The rule bites only outside that grid — an asymmetric $\delta_o \neq \delta_e$, or a different vowel on the two sides of a pause. Users who explore such conditions should be aware that the engine implements the previous-Ve anchoring, not the literal reading.

6 Why the published demonstrations are nevertheless reproduced

The audit's evidence, case by case [AUD]:

- **Equation level.** Independent implementations of Eqs. (1)–(4) agree with the engine to $\leq 5 \cdot 10^{-15}$.
- **Counterfactual CO.** Substituting the article's Table 1 coordination matrix into the engine yields identity to $\leq 10^{-15}$ on all seven parameters: the Tip-row deviation (row 1/1b) is a pure parameter substitution, not an equation change.
- **End-to-end transformation.** The `big.bi` → `bi.gbi` transformation is reconstructed from the article's equations alone to $\leq 4.4 \cdot 10^{-15}$ on six parameters, with the published gain of one period $T = 160$ ms (176 → 160 steps).
- **Locus demonstration.** With the article's positions restored — which the repository's own scripts do [R `scripts/run_locus_trough_demos.py:206-216`] — the measured locus slopes are 0.7028 (/b/), 0.4211 (/d/), 2.3352 (/g/ velar) and 0.6305 (/g/ palatal), reproduced to $\leq 5 \cdot 10^{-4}$. These are the values reported in the README headline table and in manual §10.2 [M l.959–962].
- **Word-onset rule.** Exactly zero difference at all published conditions [D5] (§5, Table 2).

README, "Key Modifications", item 4 — closing sentence (Q10, verbatim [R])

"…with the symmetric $\delta_o = \delta_e$ the value is unchanged (the 'big@bi' schwa demonstrations are preserved, and all reference reproductions remain bit-identical)."

The audit tally: 104 tests — 91 conforming, 10 documented deviations (the families of §4), 1 non-conformity (the CO Tip row), and 0 P0-severity issues [AUD]. **Consequence:** the engine faithfully applies the article's equations; every deviation is a parameter or rule choice. No engine modification is warranted; the appropriate fix is documentation — this supplement and the two corrigenda of §7.

7 Corrigenda to the manual

The audit found exactly two factually wrong statements in the manual [AUD], plus one family of wrong labels in the demonstration artifacts generated by the repository's own figure generator; the three corrections are issued below as Corrigenda 1–3. Each corrigendum gives (a) the location, (b) the current text or state quoted verbatim, (c) a ready-to-paste corrected text, and (d) the verified engine values. *Application status (October 14, 2026):* the three corrections have been applied to the manual and the figure labels regenerated from the engine; the quoted "current" texts refer to the state at commit `4a0d76290aee`.

Corrigendum 1 — coordination-matrix claim (manual §1.2, lines 225–227)

(b) Current text [M l.225–227], verbatim:

`docs/manual.tex` lines 225–227 (Q8, verbatim)

"The CO matrix matches the article's Table 1 almost exactly; the only difference is the Tip parameter c_0: -2.75 in `synthSYL/constants.py` vs -3.0 in the article."

This statement is wrong on two counts: the Tip row differs in *two* constants, not one; and no other row differs.

(c) Ready-to-paste corrected text:

Suggested replacement for manual §1.2, lines 225–227

"The CO matrix matches the article's Table 1 except for the Tip row, where two parameters differ: $\Omega = 0.5$ (article: 0) and $\Psi_1 = -2.75$ (article: -3.0) in `synthSYL/constants.py`; all other rows are identical. The deviation is inherited from the original timit-to-Maeda registry and is quantified in the Differential Supplement §4 (up to 0.70 Maeda units on Tip for /d/, /g/; zero for /b/ and /gb/)."

(d) Verified engine values. `constants.py:28` stores the Tip row as $[\Psi_1, \Omega, \Psi_2] = [-2.75,\ 0.5,\ \pi]$ [R]; the article's Table 1 gives $\Omega = 0$ and $\Psi_1 = -3$ [A]; effect up to 0.70 M.u. on Tip for /d/, /g/, zero for /b/ and /gb/ [AUD].

Corrigendum 2 — cluster-selection rule (manual §10.4, table lines 1260–1273 and rule paragraph lines 1275–1280)

(b) Current text [M l.1275–1280], verbatim:

`docs/manual.tex` lines 1275–1280 (Q9, verbatim)

"Rule: when /b/ is involved in the pair, the selection vector is $S_c = \{1, 2, 3, 6\}$ (article numbering; code $\{0, 1, 2, 5\}$ = jaw, body, dorsum, LipH); otherwise $S_c = \{1, 2, 3, 4\}$ (article numbering; code $\{0, 1, 2, 3\}$ = jaw, body, dorsum, tip). The /g/ is systematically palatal when in a cluster."

This is the *article's* rule (§3, quote in §3 above), presented as the engine's. The Demo 4 table cells are consistent with it and therefore also wrong for /dg/ and /gd/ ($\{1, 2, 3, 4\}$ is a vector the engine never returns for a cluster; the engine returns $\{1, 2, 3\}$ for both). **Explicit cell replacements**: /dg/ row [M l.1268] — S_c $\{1, 2, 3, 4\} \rightarrow \{1, 2, 3\}$ and note "no /b/ $\rightarrow$ S_c=$\{1, 2, 3, 4\}$" $\rightarrow$ "engine threshold rule (place-index sum > 2)"; /gd/ row [M l.1270] likewise.

(c) Ready-to-paste corrected text:

Suggested replacement for the manual §10.4 rule paragraph

"Engine rule (verified): the selection vector is $S_c = \{1, 2, 3, 6\}$ (code $\{0, 1, 2, 5\}$) when the sum of place indices ($b = 0$, $d = 1$, $g = 2$) is ≤ 2, and $S_c = \{1, 2, 3\}$ (code $\{0, 1, 2\}$) otherwise. Hence /bb/, /bd/, /bg/, /db/, /dd/, /gb/ use $\{1, 2, 3, 6\}$ and /dg/, /gd/, /gg/ use $\{1, 2, 3\}$; the code vector $\{0, 1, 2, 3\}$ is never produced for a cluster (it is the singleton selection of /d/ and /g/). The article's rule — $\{1, 2, 3, 6\}$ for clusters containing /b/ (e.g. /gb/), $\{1, 2, 3, 4\}$ otherwise — differs from the engine's for the four pairs /dd/, /dg/, /gd/, /gg/. The engine's rule is inherited from the 2023 lineage: both original engines implement the same threshold rule [EXT]; the article's /b/-conditional rule was never implemented in the family. See Differential Supplement §4, rows 4 and 4a."

(d) Verified engine values. Place indices b=0, d=1, g=2 [R phonology.py:133–141]; the rule selects `ART_CLUSTER[0]` = $\{0, 1, 2, 5\}$ when the sum of the two indices is ≤ 2 and `ART_CLUSTER[1]` = $\{0, 1, 2\}$ otherwise [R phonology.py:337–341; constants.py:89–92], executed on the nine pairs for this supplement: {0,1,2,5} for bb, bd, bg, db, dd, gb; {0,1,2} for dg, gd, gg (article numbering: $\{1, 2, 3, 6\}$ and $\{1, 2, 3\}$) [R]. The article's rule is quoted verbatim in §3 [A §3]. The affected utterances of the extended simulations are `adga` and `agda` (see Row 4b in §4).

Corrigendum 3 — demonstration-figure labels (generator `run_article_demos.py`; manual embeds at lines 1284 and 1291)

(a) Location. The demonstration figure generator `scripts/run_article_demos.py` hardcodes the article's labels in `CLUSTER_INFO` (lines 91–98): the /dg/ and /gd/ entries carry `"Sc": "{1,2,3,4}"` with note `"no /b/ -> Sc={1,2,3,4}"` (verbatim excerpt in the brief's Annex A). These strings flow into (i) the console log (line 310), (ii) the panel titles of `cluster_params.png` (line 356), and (iii) the panel titles of `cluster_polar.png` (line 416). The manual embeds both PNGs ([M l.1284] and [M l.1291]), so the printed Demo 4 section inherits the mislabels.

(b) Current state. The synthesis that produced those very panels ran with $S_c = \{1, 2, 3\}$ for /dg/ and /gd/ (threshold rule; §4, row 4, verified by execution). The published artifacts therefore display $S_c = \{1, 2, 3, 4\}$ — a vector the engine never returns for a cluster — next to trajectories that are faithful to the engine: a label/engine inconsistency inside the published Demo 4 artifacts. Only the labels are wrong. The spectrogram titles carry no S_c label, and Demo 1 (/ibia/) is unaffected.

(c) Correction — two coordinated parts. *Generator (maintainer action).* Option B (recommended): derive the labels at runtime from `synthSYL.phonology.cluster_art_params`, converting code indices to article numbering by +1, so the displayed labels can never drift from the engine again; keep a static note field for prose only. Option A (minimal): edit the two entries to `"Sc": "{1,2,3}"` with note "engine threshold rule (place-index sum > 2)". Then rerun the generator, propagate the regenerated `cluster_params.png` and `cluster_polar.png` into `docs/figures/demo5/`, and rebuild the manual PDF. *Ready-to-paste manual note (immediate documentation fix), to insert under the Demo 4 figures:*

Suggested note under the manual Demo 4 figures

"Labels note. In the figures above, the S_c annotations for /dg/ and /gd/ ($S_c = \{1, 2, 3, 4\}$) are legacy labels: the synthesis runs with $S_c = \{1, 2, 3\}$ (threshold rule on the place-index sum; see the Differential Supplement, §4 row 4, and Corrigendum 2). The trajectories are unaffected; corrected figures are regenerated by `scripts/run_article_demos.py`."

(d) Verified engine values. /dg/ and /gd/ $\rightarrow$ code $[0, 1, 2] \equiv \{1, 2, 3\}$ (article numbering); the four panels /bd/, /bg/, /db/, /gb/ already display the correct $\{1, 2, 3, 6\}$. Panel by panel:

Panel	Displayed label	Engine value	Action
/bd/, /bg/, /db/, /gb/	$S_c = \{1, 2, 3, 6\}$	$\{1, 2, 3, 6\}$	none
/dg/	$S_c = \{1, 2, 3, 4\}$ + "no /b/" note	$\{1, 2, 3\}$	relabel
/gd/	$S_c = \{1, 2, 3, 4\}$ + "no /b/" note	$\{1, 2, 3\}$	relabel

8 Version and provenance statement

The following block is ready to paste into the README and the manual preface:

Version and provenance statement (ready to paste)

Version and provenance. The demonstrations in this repository were produced with the Python engine synthSYL reworked in 2026 within big.bi-supplement (commit `4a0d76290aee`, DOI 10.5281/zenodo.23239483), which extends the author's timit-to-Maeda package. The 2023 article (arXiv:2307.02299) reports demonstrations produced with the author's MATLAB implementation of the same model. The registry parameters are inherited from the engine and recalibrated by ear; they do not all coincide with the values published in the 2023 article — the complete differential is catalogued in the Implementation Differential and Provenance Supplement (`docs/differential_supplement.pdf`, §4), and none of the differences is an equation-level change. Two choices are dated author rulings: the word-onset anchoring rule (2026-10-06) and VALRECT = 1.10 (2026-10-07).

9 Reproducibility

Environment. Python ≥ 3.10 (the audit ran 3.12) with numpy, scipy, matplotlib and pytest; repository at commit `4a0d76290aee`; MIT license [R]. The commands below follow the audit workspace [AUD; D5]; steps 3–5 use the scripts `exp_d5_onset_rule.py` and `fig_d5.py` provided there (on Windows, replace `venv/bin/` with `venv\Scripts\` and set `BIGBI_REPO` with `set`):

```
# 1. environment
python -m venv venv && venv/bin/pip install numpy scipy matplotlib pytest
# 2. repository at the pinned commit
git clone https://github.com/FBerthommier/big.bi-supplement
cd big.bi-supplement && git checkout 4a0d76290aee
# 3. difference-5 experiment (~10 s)   [script: audit workspace]
BIGBI_REPO=/path/to/repo python exp_d5_onset_rule.py
#    -> d5_results.json + d5_<case>_{code,lit}.csv (14 CSVs)
# 4. figures
python fig_d5.py   # -> fig_d5_bibi.png, fig_d5_bigbi.png, fig_d5_synthesis.png
# 5. full 104-test audit suite (~10 min)
cd tests/compliance && BIGBI_REPO=/path/to/repo python -m pytest . -q
```

Artifact inventory. Difference-5: `d5_results.json`, 14 per-case CSVs (engine vs literal rule), three figures (`fig_d5_bibi.png`, `fig_d5_bigbi.png`, `fig_d5_synthesis.png`) [D5]. Audit: the 104-test matrix (`compliance_report.md`), the engine-vs-article correspondence table (`annexe_correspondance_synthsyl_article.md`), and the 13-page difference-5 report (`difference5_effect_audit.pdf`) [AUD].

Selector probe. Every selector claim of this document is re-derivable from the repository alone. Run from any directory with the repository root in `sys.path`:

```
import sys
```

```
sys.path.insert(0, "<repository root>")
from synthSYL.phonology import cluster_art_params

def article_rule(pair):          # article S3, p. 4
    return {1, 2, 3, 6} if "b" in pair else {1, 2, 3, 4}

for p in ["bb", "bd", "bg", "db", "dd", "dg", "gb", "gd", "gg"]:
    code = cluster_art_params(list(p))          # 0-based code indices
    art  = sorted(i + 1 for i in code)          # article numbering (+1)
    ref  = sorted(article_rule(p))
    print(f"/{p}/  engine {art}  article {ref}  "
          f"{'conform' if art == ref else 'DIVERGES'}")
```

Expected output: `conform` for bb, bd, bg, db, gb (engine $\{1,2,3,6\}$); `DIVERGES` for dd (engine $\{1,2,3,6\}$) and for dg, gd, gg (engine $\{1,2,3\}$); the article rule answers $\{1,2,3,4\}$ for these four. Place-index sums (b=0, d=1, g=2): bb 0; bd 1; bg 2; db 1; dd 2; dg 3; gb 2; gd 3; gg 4 — the threshold is sum > 2.

References

[1] F. Berthommier, "Why can big.bi be changed to bi.gbi? A mathematical model of syllabification and articulatory synthesis", arXiv:2307.02299 [eess.AS], July 2023 (Interspeech 2023). Cited as [A].

[2] F. Berthommier, *big.bi-supplement* repository, commit `4a0d76290aee`, MIT license, DOI 10.5281/zenodo.23239483, 2026. https://github.com/FBerthommier/big.bi-supplement. Cited as [R].

[3] F. Berthommier, *Didactic Manual: big.bi → bi.gbi articulatory simulation*, `docs/manual.pdf`, 26 pages, at the same commit. Cited as [M].

[4] 104-test compliance audit report, `compliance_report.md`, October 2026. Cited as [AUD].

[5] Difference-5 effect report, `difference5_effect_audit.pdf`, 13 pages, October 2026. Cited as [D5].

[6] Anonymous, "Comparative audit of consonant selection vectors $C \in$ {b, d, g} in Timit-to-Maeda and Syllable_Synthesis," technical synthesis for verification of reasoning, 6 pp., October 13, 2026 (file `selectors_CC_comparison.pdf`). Cited as [EXT].